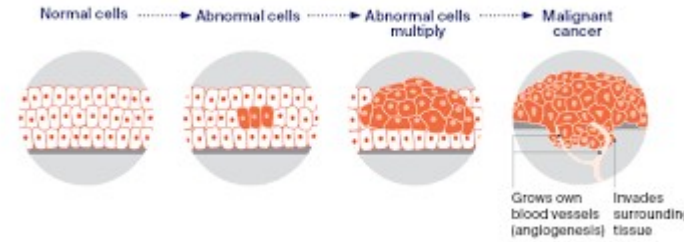


ANTI-CANCER AGENTS

Solomon A.

1

WHAT IS CANCER??



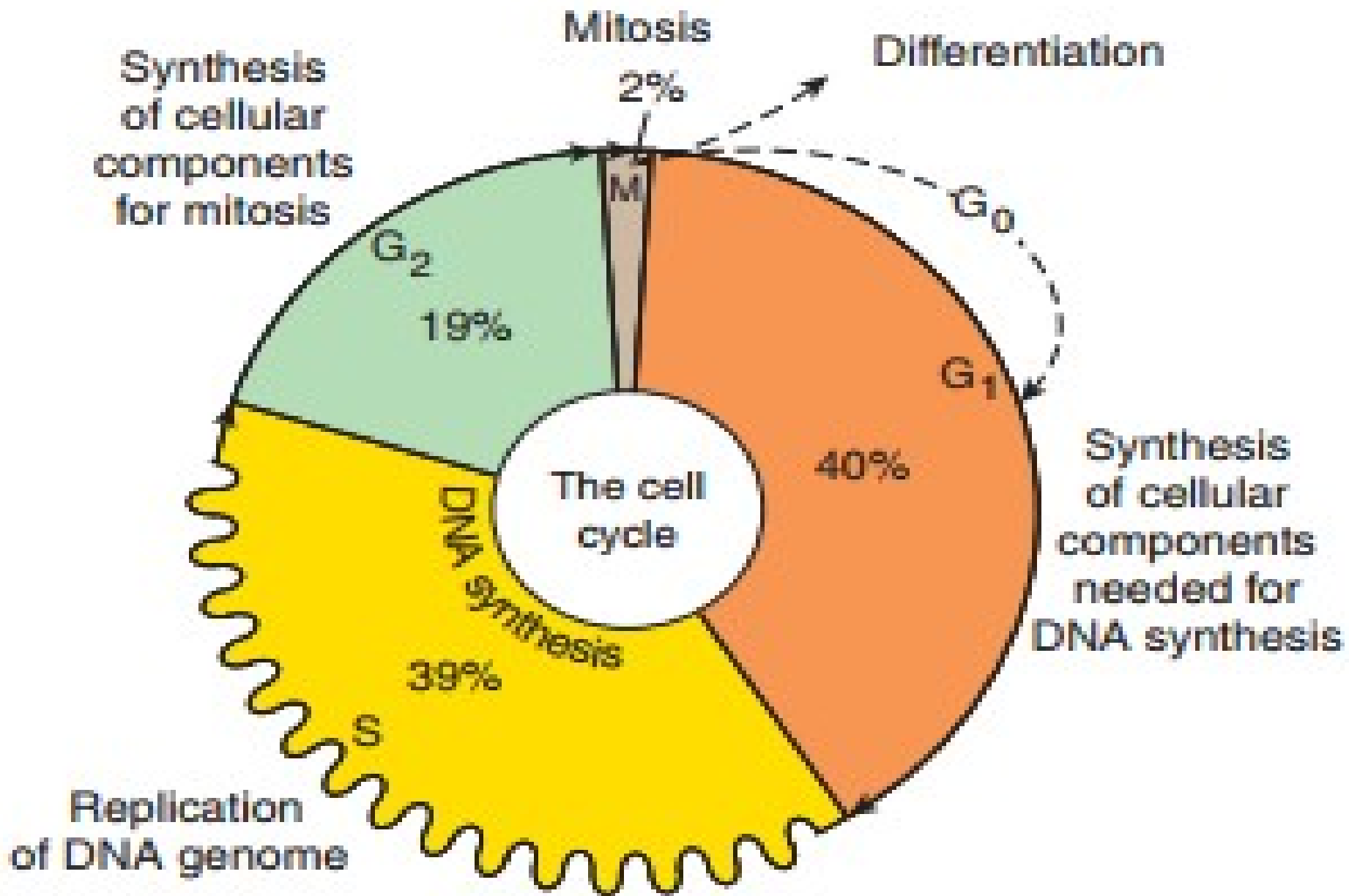
- Cancer is a disease characterized by a loss in the normal control mechanisms that govern cell survival, proliferation, and differentiation.
- Cancer is the uncontrolled development of cells.
- Oncogenic addiction (Achilles heel)
 - ✓ Tumor stem cells
 - ✓ Genetic instability

CAUSES OF CANCER

- Behavior of specific types of cancer are related to multiple factors:
 - Sex
 - Age
 - Race
 - Genetic predisposition
 - Exposure to environmental carcinogens
- Several **viruses** have been implicated in the etiology of various human cancers
- **Oncogenes** and **tumor suppressor genes**

PRINCIPLE

- Earlier diagnosis might lead to increased cure rates with such local treatment.
- Chemotherapy alone cures $<10\%$ at an advanced stage
- Chemotherapy is presently used in three main clinical settings:
 - 1) Primary induction treatment
 - 2) Neoadjuvant treatment
 - 3) Adjuvant treatment to local methods of treatment

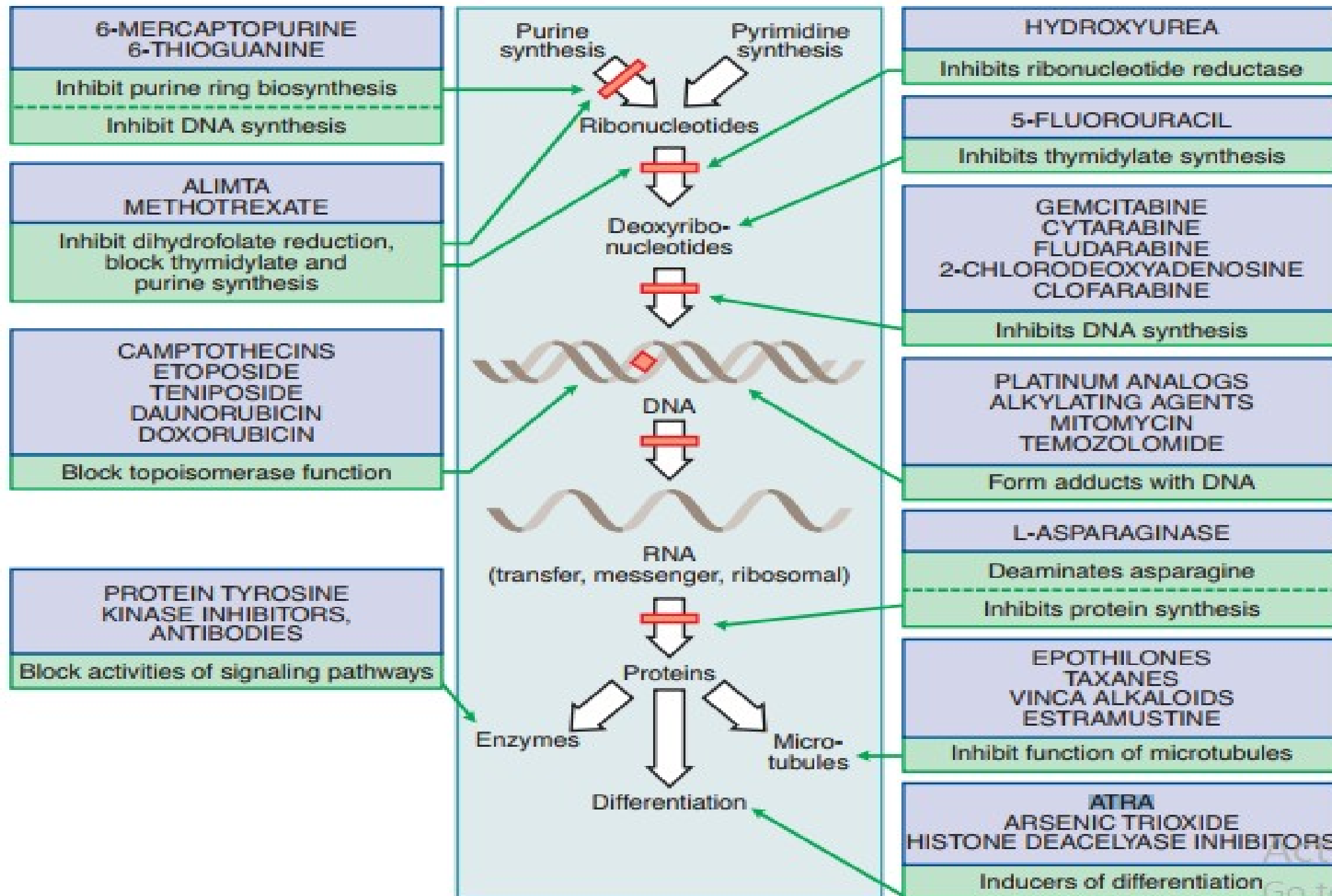


THE ROLE OF DRUG COMBINATIONS

- Single drugs at clinically tolerable doses have been unable to cure cancer.
- Drug interaction considers:
 - ✓ Efficacy
 - ✓ Toxicity
 - ✓ Optimum scheduling
 - ✓ Mechanism of interaction

RESISTANCE

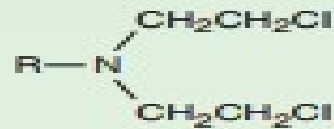
- A fundamental issue in cancer chemotherapy is the development of cellular drug resistance.
- Some tumor types exhibit *primary* resistance, ie, absence of response on the first exposure, to currently available agents.



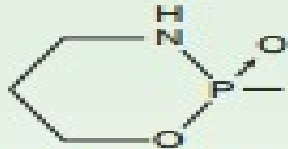
BASIC PHARMACOLOGY OF CANCER CHEMOTHERAPEUTIC DRUGS

ALKYLATING AGENTS

Bis(chloroethyl)amines



Where R is:



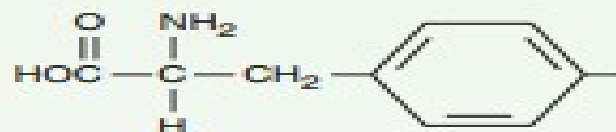
Cyclophosphamide



Mechlorethamine

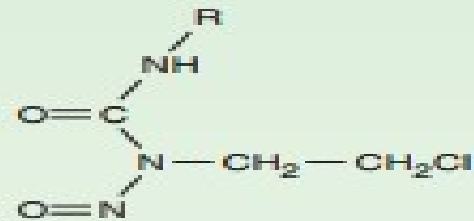


Chlorambucil



Melphalan

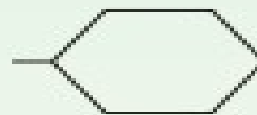
Nitrosoureas



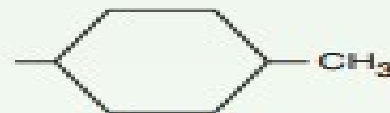
Where R is:



BCNU
(carmustine)

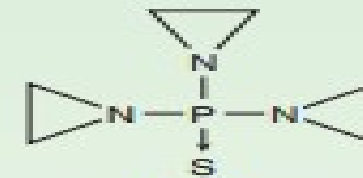


CCNU
(lomustine)

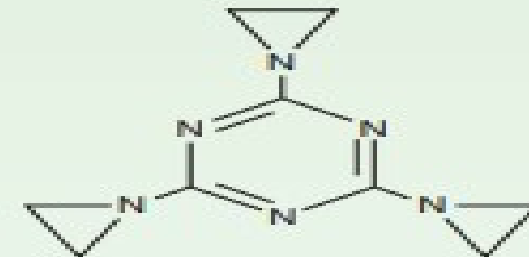


Methyl-CCNU
(semustine)

Aziridines

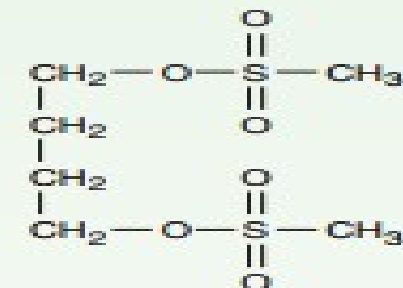


Thiotepa



Triethylenemelamine

Alkylsulfonate

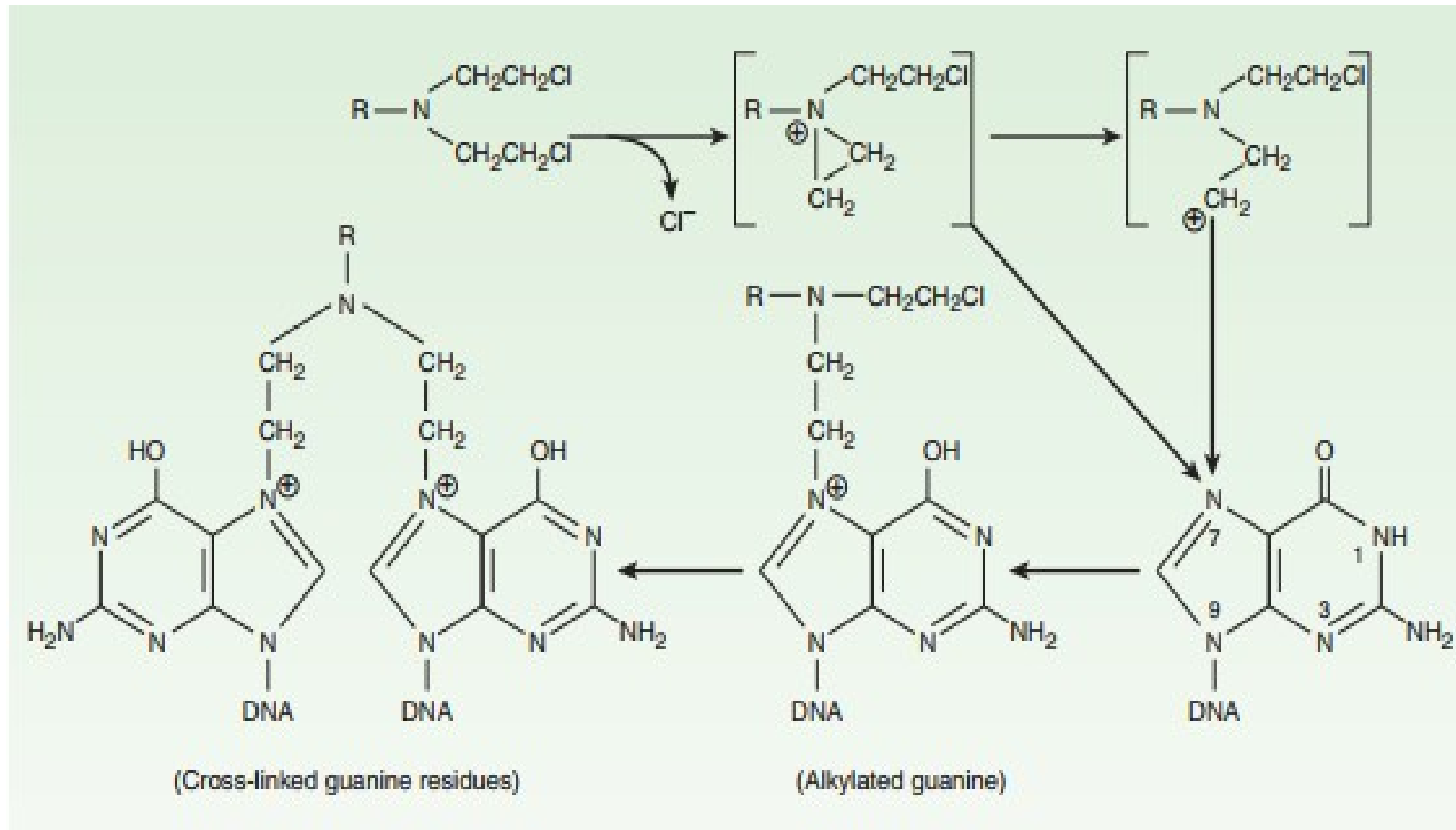


Busulfan

CONT...

- At present , six major types of alkylating agents are used in the chemotherapy of neoplastic diseases:
 - ✓ Nitrogen mustards
 - ✓ Ethyleneimines
 - ✓ Alkyl sulfonates
 - ✓ Nitrosoureas
 - ✓ The triazenes
 - ✓ DNA methylating drugs including procarbazine, temozolmide and dacarbazine

HOW THEY WORK??

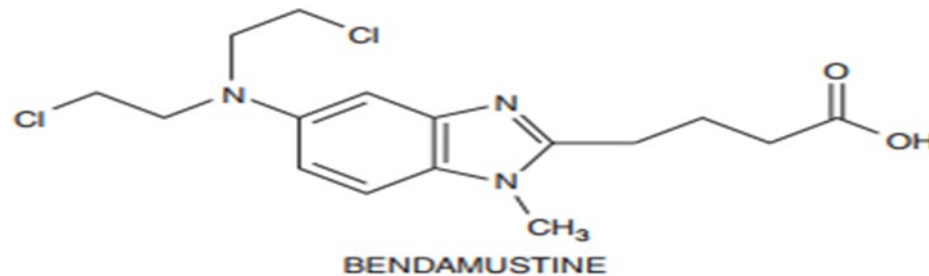


CONT...

- Transfer of their alkyl groups to various cellular constituents.
 - Alkylations of DNA within the nucleus
 - Carbamoylation of lysine residues of proteins through formation of isocyanates
 - ✓ N7 position of guanine
 - ✓ Not cell cycle specific
- Cancerous cell can develop resistance

CONT...

- Nausea and vomiting can be a serious issue with a number of these agent
- Vesicants
- Carcinogenic



NONCLASSIC ALKYLATING AGENTS

- Procarbazine
- Dacarbazine
- Bendamustine

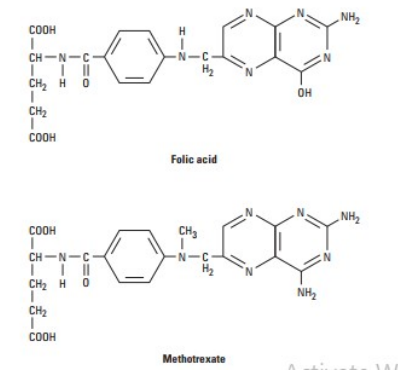
PLATINUM ANALOGS

- Three platinum analogs are currently used in clinical practice:
- Cisplatin: broad range of solid tumors/ urinary excretion
- Carboplatin: broad-spectrum activity/ less nephrotoxicity /easy to administer
- Oxaliplatin: in CRC/ Non- cross-resistant /FOLFOX regimen, for metastatic colorectal cancer

TOXICITY OF ALKYLATING AGENTS

- Bone marrow suppression
- Mucosal toxicity
- Neurotoxicity
- Reproductive toxicity
- Leukemogenesis

ANTIMETABOLITES



- Important both conceptually and clinically

Methotrexate

- Folic acid analog
- Targets dihydrofolate reductase (DHFR)
- ✓ Interferes with the formation of DNA, RNA, and key cellular proteins
- ✓ MTX polyglutamates are selectively retained within cancer cells

CONT...

- Resistance to MTX has been attributed to
 - (1) Decreased drug transport
 - (2) Decreased MTX polyglutamates,
 - (3) Increased DHFR
 - (4) Altered DHFR protein

- Care must also be taken when MTX is used in the presence of drugs such as aspirin, nonsteroidal anti-inflammatory agents, penicillin, and cephalosporins
- Leucovorin rescue

ADR MTX

- ❑ Nausea
- ❑ Mucositis
- ❑ Diarrhea
- ❑ Bone Marrow Depression

PEMETREXED

- S phase of the cell cycle.
- Targets DHFR and enzymes involved in de novo purine nucleotide biosynthesis
- TS*
- Mesothelioma
- as a single agent in the second-line therapy of non-small cell lung cancer
- in combination with cisplatin for the first-line treatment of non-small cell lung cancer.
- ADR
 - myelosuppression, skin rash, mucositis, diarrhea, fatigue, and hand-foot syndrome

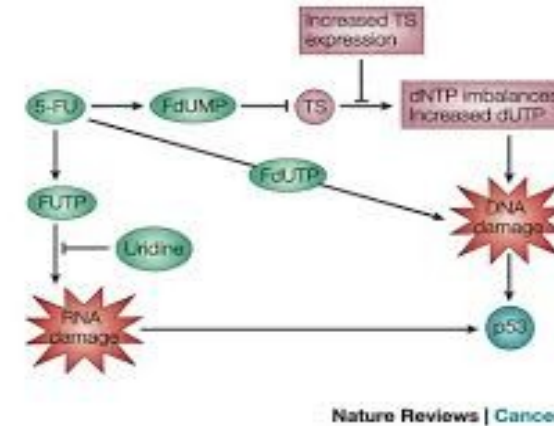
PRALATREXATE

- 10-deaza-aminopterin antifolate analog
- It inhibits DHFR, inhibits enzymes involved in de novo purine nucleotide biosynthesis, and also inhibits thymidylate synthase.
- Originally developed for non-small lung cancer, it is presently approved for use in the treatment of relapsed or refractory peripheral T-cell lymphoma.

5-FLUOROURACIL

○ FLUOROPYRIMIDINES

- Requires activation via a complex series of enzymatic reactions to ribosyl and deoxyribosyl nucleotide metabolites
- 5-fluoro-2-deoxyuridine-5-monophosphate (FdUMP), forms a covalently bound ternary complex with the enzyme thymidylate synthase and the reduced folate 5,10 methylenetetrahydrofolate
- 5-fluorouridine-5-triphosphate (FUTP)
- 5-fluorodeoxyuridine-5'-triphosphate(FdUTP)
- Dihydropyrimidine dehydrogenase (DPD)



- Capecitabine
- Cytarabine
- Gemcitabine

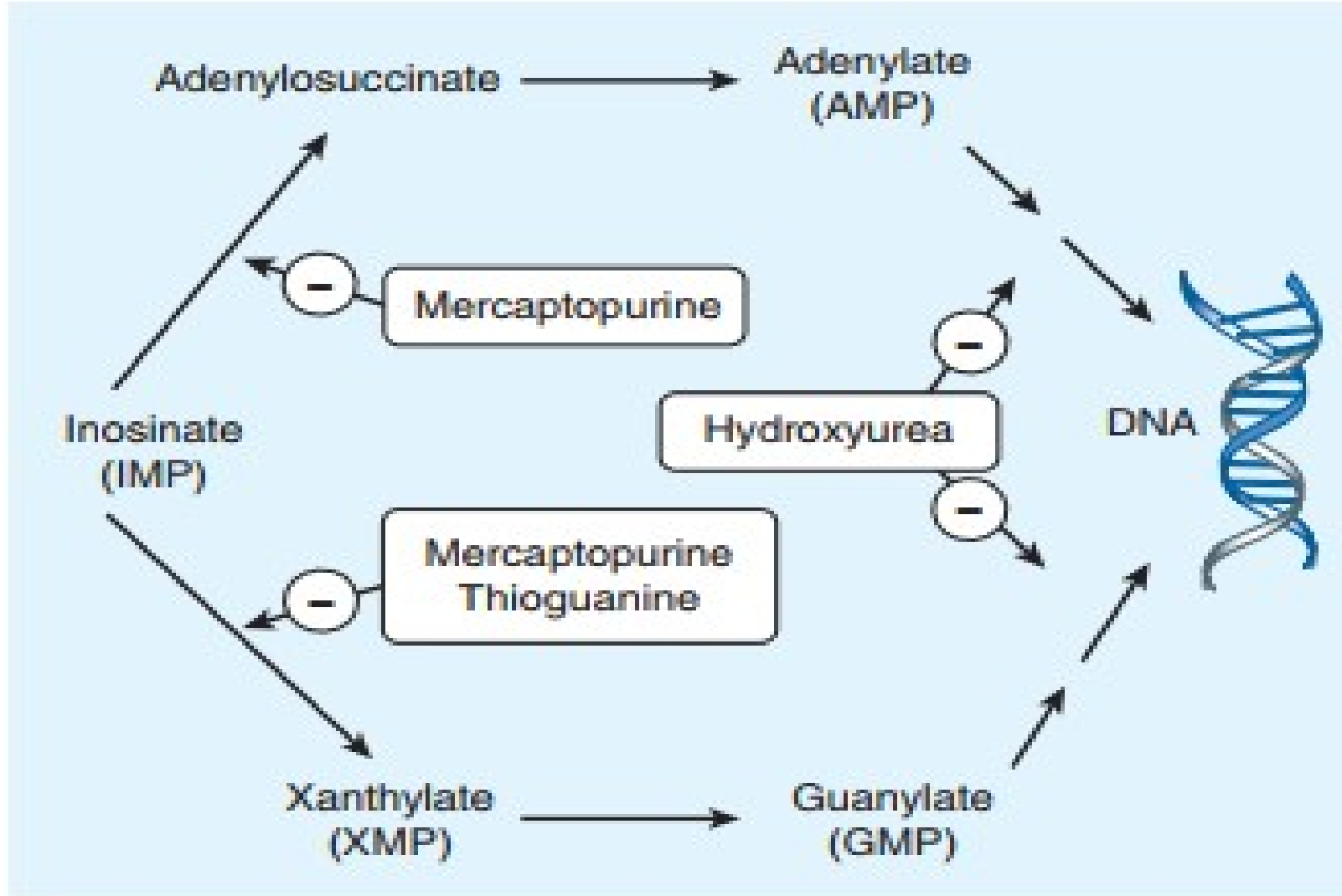
PURINE ANTAGONISTS

- 6-Thiopurines
- 6-Mercaptopurine (6-MP) was the first of the thiopurine analogs found to be effective in cancer therapy
- 6-MP is inactive
- Metabolized by hypoxanthine-guanine phosphoribosyl transferase (HGPRT) to form the monophosphate nucleotide 6-thioinosinic acid
- Treatment of childhood acute leukemia
- 6-MP gives 6-thiouric acid by xanthine oxidase
- 6-TG undergoes deamination

QUIZ 15 MIN

- 1) Discuss the way cancer cells can develop resistance against anticancer
- 2) List the possible anticancer mechanism of actions
- 3) What does it mean when we say:
 - A) Oncogene vs tumor suppressor gene
 - B) Cell cycle specific and non-specific agents
- 4) Try to reflect on the use of mesna

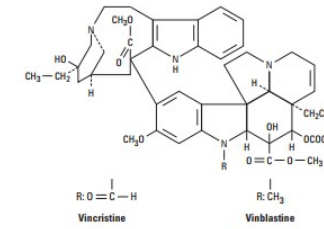
Note: you can google but write on your way or write from the perspective we discussed here



CONT....

- **Fludarabine**
Cladribine

NATURAL PRODUCT CANCER CHEMOTHERAPY DRUGS



VINCA ALKALOIDS

Vinblastine

- Vinblastine is an alkaloid derived from the periwinkle plant *Vinca rosea*
- Inhibits tubulin polymerization
- Results in mitotic arrest in metaphase
- It has clinical activity in the treatment of Hodgkin's and non-Hodgkin's lymphomas, breast cancer, and germ cell cancer.

CONT...

Vincristine

- ✓ Vincristine has a strikingly different spectrum of clinical activity and safety profile
- ✓ Vincristine has been effectively combined with prednisone for remission induction in acute lymphoblastic leukemia in children

Vinorelbine

OTHERS...

- Paclitaxel
- Docetaxel
- Cabazitaxel
- Etoposide
- Topotecan and irinotecan

ANTITUMOR ANTIBIOTICS

33

ANTHRACYCLINES

- isolated from *Streptomyces peucetius* var *caesius*
- major mechanisms:
 - (1) inhibition of topoisomerase II
 - (2) high-affinity binding to DNA through intercalation, with consequent blockade of the synthesis of DNA and RNA, and DNA strand scission;
 - (3) generation of semiquinone free radicals and oxygen free radicals through an iron-dependent, enzyme-mediated reductive process;
 - (4) binding to cellular membranes to alter fluidity and ion transport.

CONT...

Doxorubicin

- important anticancer drugs in clinical practice, in cancers of the breast, endometrium, ovary, testicle, thyroid, stomach, bladder, liver, and lung; in soft tissue sarcomas; and in several childhood cancers
- It also has clinical activity in hematologic malignancies, (acute lymphoblastic leukemia, multiple myeloma, and Hodgkin's and non-Hodgkin's lymphomas)

CONT...

- Daunorubicin
- Idarubicin
- Epirubicin
- Mitoxantrone

MITOMYCIN

- Isolated from *Streptomyces caespitosus*
- Generate an alkylating agent that cross-links DNA.
- Active in all phases of the cell cycle
- The best available drug for use in combination with radiation therapy to attack hypoxic tumor cells.
- Used in squamous cell carcinoma of the anus, cervix and for breast, gastric, and pancreatic cancer.
- The common toxicities: Hemolytic-uremic syndrome, manifested as microangiopathic hemolytic anemia, thrombocytopenia, and renal failure, as well as occasional instances of interstitial pneumonitis

BLEOMYCIN

- Bleomycin is a small peptide
- contains a DNA-binding region and an iron-binding domain at opposite ends of the molecule
- results in single- and double strand breaks following free radical formation, and inhibition of DNA biosynthesis.
- Bleomycin is a cell cycle-specific drug that causes accumulation of cells in the G2 phase of the cell cycle.
- Indicated for the treatment of Hodgkin's and non-Hodgkin's lymphomas, germ cell tumor, head and neck cancer, and squamous cell cancer of the skin, cervix, and vulva.
- One advantage of this agent is that it can be given subcutaneously, intramuscularly, or intravenously.

MISCELLANEOUS ANTICANCER DRUGS

- **IMATINIB, DASATINIB, & NILOTINIB**
- **Cetuximab & Panitumumab**
- **Gefitinib & Erlotinib**
- **Bevacizumab, Sorafenib, Sunitinib, & Pazopanib**
- **ASPARAGINASE**